

Mahindra University Hyderabad
 École Centrale School of Engineering
 Minor-I

Program: B. Tech.

Branch: CSE/ECE/AI Year: III
 Subject: Operating System CS3102

Semester: I

Date: 13/09/2025

Time Duration: 1.5 Hours

Start Time: 02:00 PM

Max. Marks: 30

Instructions:

1. Please write the answer in sequence as given in the question paper.
2. Attempt all questions.

Q. No.	Questions	Marks	CO	BL	PI																								
1	Explain the booting process step-by-step.	5	CO-1	L-1																									
2	What is the dual mode of the CPU? Why CPU execute processes in dual mode? Explain with an example.	5	CO-1	L-1																									
3	Find the average turnaround time, average waiting time, average response time, scheduling length, and throughput for the following data using a non-preemptive priority scheduling algorithm. Consider 1 as the highest priority <table><tr><th>Process-id</th><th>Arrival time</th><th>Burst Time</th><th>Priority</th></tr><tr><td>P1</td><td>0</td><td>3</td><td>5</td></tr><tr><td>P2</td><td>2</td><td>2</td><td>2</td></tr><tr><td>P3</td><td>2.5</td><td>6</td><td>3</td></tr><tr><td>P4</td><td>4</td><td>1</td><td>1</td></tr><tr><td>P5</td><td>5</td><td>2</td><td>4</td></tr></table>	Process-id	Arrival time	Burst Time	Priority	P1	0	3	5	P2	2	2	2	P3	2.5	6	3	P4	4	1	1	P5	5	2	4	5	CO-2	L-2	
Process-id	Arrival time	Burst Time	Priority																										
P1	0	3	5																										
P2	2	2	2																										
P3	2.5	6	3																										
P4	4	1	1																										
P5	5	2	4																										
4	a) Draw and explain the life cycle of a process in an Operating System. - Clearly label all the process states. - Show the transitions between states. b) Briefly describe the role of all states.	2+3	CO2	L-1																									
5	Four processes, P1, P2, P3, and P4 arrive at time 0 with total execution times of 20, 40, 60, and 20 units, respectively. Each process executes in the following pattern: - First 25% of its time in I/O, - Next 60% in CPU (computation), - Final 15% in I/O.	10	CO2	L-3																									

	The operating system uses a Shortest Remaining Compute Time First (SRCTF) scheduling policy. A new process is scheduled only when the currently running process blocks for I/O or completes its CPU burst. Assume all I/O operations can overlap completely. a) Draw the CPU execution timeline for all four processes. (4 Marks) b) Calculate the percentage of time the CPU remains idle. (3 Marks) c) Find the throughput of the system. (3 Marks)				
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